

Merged Engineering Change

PR #284 / Validate upload size before reading request body

MERGED

main / commit 7c91e2a / reviewed by Platform Engineering

Author	Elias Morgan, Senior Backend Engineer	Risk	Medium
Service	Upload API	Ticket	ENG-913
Opened	2026-06-19	Merged	2026-06-20

Problem

Problem: The Upload API began reading large request bodies before checking the configured tenant limit. Oversized requests consumed memory and sometimes timed out instead of returning a clear HTTP 413 response.

Root cause and decision

Root cause: Validation happened after request-body allocation, and the boundary check was inconsistent at exactly 200 MiB.

Reasoning: The expert chose a small validation function at the request boundary because rejecting an invalid Content-Length before reading the body was safer, easier to test, and reversible. The change preserved the documented rule that exactly 200 MiB is accepted.

Completed implementation

1. Added a pure `validate_upload_size` function with explicit handling for missing, negative, accepted, and oversized lengths.
2. Used a strict greater-than comparison so a file exactly at the configured limit remains valid.
3. Returned HTTP 411 for missing Content-Length, HTTP 400 for an invalid negative value, and HTTP 413 with a customer-readable limit message for oversized requests.
4. Added five regression tests covering below-limit, exact-limit, above-limit, missing-header, and invalid-configuration behavior.
5. Ran the focused test file and the upload-service regression suite before merge.

Files changed

`upload_handler.py` - validation and response behavior

`test_upload_handler.py` - five boundary and failure-path tests

Verification

Check	Result
Focused tests	5 passed
Upload-service regression suite	48 passed
Static analysis	Passed
Canary: valid 150 MiB upload	Succeeded
Canary: 201 MiB upload	HTTP 413 with documented message

Outcome

Outcome: PR #284 merged after review. Oversized requests are now rejected before body allocation, valid requests at the exact limit succeed, the canary remained healthy for 30 minutes, and no upload-memory alerts occurred after deployment.

Reusable judgment

For request-boundary bugs, define the exact accepted and rejected values first, implement the narrowest validation point, and keep regression tests for both sides of the boundary.